

Samadhini Perera

Computer Engineering Undergraduate | Internship Applicant

Kurunegala District, Sri Lanka | e21289@eng.pdn.ac.lk | github.com/samadhini perera | linkedin.com/in/samadhini-perera-263a842b8

Objective

Third-year Computer Engineering undergraduate (University of Peradeniya) seeking an internship across software, AI/ML, and IoT. Builds full-stack platforms, embedded/IoT systems, and applied research and looking to bring that range to a collaborative engineering team.

Education

BSc Eng. (Hons) in Computer Engineering

University of Peradeniya | Jun 2023 – Present (Expected 2027)
3rd Year; Data Structures & Algorithms, Software Engineering, Networks, Databases, OS, Embedded Systems, ML

Technical Skills

Languages	Python	Java	C	JavaScript	SQL
Web	React	Node.js	Express	REST APIs	Tailwind
Data / Cloud	PostgreSQL	Supabase	MongoDB	AWS EC2	Docker
Embedded / IoT	ESP32	Arduino	MQTT	Sensor Integration	
AI / ML	Machine Learning	LLMs	Prompt Eng.	Agent-Based Modelling	
Other	Git	ArcGIS	AutoCAD	Excel	

Projects

Event Management Dashboard <i>React · Node.js · Express · Supabase · Vercel</i> Full-stack event platform: role-based auth, live attendance tracking, ratings, analytics dashboard 20+ REST endpoints, 15+ React components; live at peraverse.vercel.app	Nov 2024 – Nov 2025
ABM Flood Simulation — Kelani River Basin <i>Python · Mesa · SciPy · Plotly Dash · ArcGIS</i> Custom 3D MeshSpace (Delaunay triangulation) extending Mesa for terrain-based flood routing O(N) complexity, 100 agents in 0.06s; accepted for IEEE ICEEET 2026 (see Publications)	2025 – 2026
HYDRA — Smart Adaptive Traffic Control System <i>ESP32 · MQTT · Node.js · AWS · React</i> Sensor-driven adaptive signal system; led hardware integration, cloud deployment & ultrasonic sensing Field-validated: 86.2% average wait-time reduction vs. fixed-timer control (live at HYDRA.app)	2026
Green Node — AI Greenhouse IoT Orchestrator <i>IoT · Edge Computing · MQTT</i> Edge-first IoT hub unifying greenhouse sensors/actuators via MQTT with AI-driven automation Top 10 (Round 1) — Startup Spark AgriTech national competition	2026 – In Progress
Task Manager App <i>Node.js · Express · React</i> Full-stack CRUD task manager with RESTful API design	2025

Research Publication

Terrain-Aware ABM with 3D Delaunay Mesh Visualization in MESA: Kithulgala Sub-Basin, Sri Lanka IEEE ICEEET 2026 Designed the MeshSpace class & hydrodynamic module (Manning depth, velocity, Froude number); built the Plotly Dash dashboard	Accepted for Presentation
---	---------------------------

Experience

Casual Instructor — Department of Computer Engineering University of Peradeniya Instruct Networks, Network Analysis & Programming Methodology; grade labs/projects, mentor students	Nov 2024 – Present
Vice President, Public Relations — Toastmasters International Peradeniya University Toastmasters Club Lead club marketing: flyers, promotion videos, social media & website — builds leadership and creative/marketing skills	2026/27 Term

Additional Information

- Certifications: ML for Beginners, Prompt Engineering (Sololearn) · Agile PM, Front-End Web Dev (Moratuwa) · HackerRank Achievement
- Languages: Sinhala, English
- Interests: Yoga, Badminton